Housing Market Analysis using Tableau

Project Documentation

Group Project

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Course: Data Visualization / Tableau
Tool Used: Tableau, Flask, HTML
Dataset: Transformed_Housing_Data2.csv

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1. Problem Statement

The housing market contains many factors that influence property prices, such as house age, renovation status, and structural features like bedrooms, bathrooms, and floors. However, it can be difficult to understand these relationships using raw data alone.

The objective of this project is to analyze housing market trends and identify patterns affecting house prices using data visualization techniques in Tableau.

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2. Objectives of the Project

The main objectives of this project are:

- To analyze housing market data using Tableau.
- To understand the relationship between house price and house features.
- To visualize housing trends using interactive dashboards.
- To explore the impact of renovations on property values.
- To present insights using dashboards and storyboards.

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3. Data Collection

The dataset used in this project was obtained from a housing market dataset containing property details.

The dataset contains **21,609 records** and includes the following attributes:

- Sale Price
- Age of House (in Years)
- Years Since Renovation
- Number of Bedrooms
- Number of Bathrooms
- Number of Floors
- Basement Area
- Living Area
- Lot Area
- Renovation Status

The dataset was stored in CSV format and imported into Tableau for analysis.

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4. Data Cleaning and Preparation

Before creating visualizations, the dataset was reviewed and cleaned.

The following steps were performed:

1. **Data Review**
    - Checked column names and data types.
    - Verified numeric and categorical variables.
2. **Handling Missing Values**
    - Ensured that important fields such as sale price and house age did not contain missing values.
3. **Filtering Data**
    - Removed unnecessary columns.
    - Focused only on relevant attributes for the analysis.
4. **Field Renaming**
    - Renamed unclear fields for better understanding.
5. **Calculated Fields**

- Created calculated metrics such as:
  - Years since renovation
  - Aggregated sales metrics.

After cleaning, the dataset became easier to analyze and visualize.

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5. Data Visualization and Dashboard Creation

Multiple visualizations were created in Tableau to analyze the housing data.

5.1 Count of Transformed Housing Data

This visualization shows the **total number of housing records** in the dataset, which is 21,609.

5.2 Average Sale Price

This chart displays the **average housing price** across the dataset, helping identify general pricing trends.

5.3 Area of House from Basement (in Sqft)

This visualization represents the **total basement area of houses** in the dataset.

5.4 Total Sales by Years Since Renovation

This chart analyzes how **recent renovations affect property sales**.

It shows the relationship between sale prices and the number of years since the last renovation.

5.5 Distribution of House Age by Renovation Status

A pie chart was used to display how house age is distributed between renovated and non-renovated properties.

This helps identify which houses are more likely to be renovated.

5.6 House Age Distribution by Bathrooms, Bedrooms, and Floors

This bar chart compares house age with structural features such as:

- Number of bathrooms
- Number of bedrooms
- Number of floors

It helps identify how house design features relate to property age.

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6. Dashboard Development

A **comprehensive Tableau dashboard** was created to combine all visualizations in one view.

The dashboard includes:

- Count of Houses
- Average Sale Price
- Total Basement Area
- Total Sales by Years Since Renovation
- House Age Distribution
- Structural Feature Comparison

The dashboard provides a clear and interactive way to explore housing market trends.

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7. Storyboard Creation

A Tableau **Story** was created to present the analysis step by step.

The story includes the following stages:

1. Overview of housing data
2. Impact of renovation on property values
3. Distribution of house age
4. Relationship between house features and house age

The storyboard helps communicate insights in a logical narrative format.

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8. Web Application Development

A simple web application was created using **Flask and HTML**.

The purpose of the web application is to embed the Tableau dashboard and story inside a custom web interface.

The web application includes the following pages:

- Home page
- About page
- Dashboard page
- Story page

Users can navigate through the pages to view housing market visualizations.

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9. Results and Insights

From the analysis, several insights were discovered:

- Renovated houses tend to have higher sale values.
- The age of houses significantly influences renovation patterns.
- Houses with more bedrooms and bathrooms tend to have higher prices.
- Basement area contributes to overall property value.

These insights can help understand housing market behavior and assist in making informed decisions.

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10. Tools and Technologies Used

The following tools were used in the project:

- Tableau for data visualization
- Python Flask for web application development
- HTML and CSS for web interface design
- CSV dataset for housing data

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11. Conclusion

This project successfully analyzed housing market data using Tableau visualizations and dashboards. By combining interactive charts, dashboards, and a web application

interface, the project provides meaningful insights into housing trends and property features.

The visualizations help users easily explore housing data and understand the factors influencing property prices.

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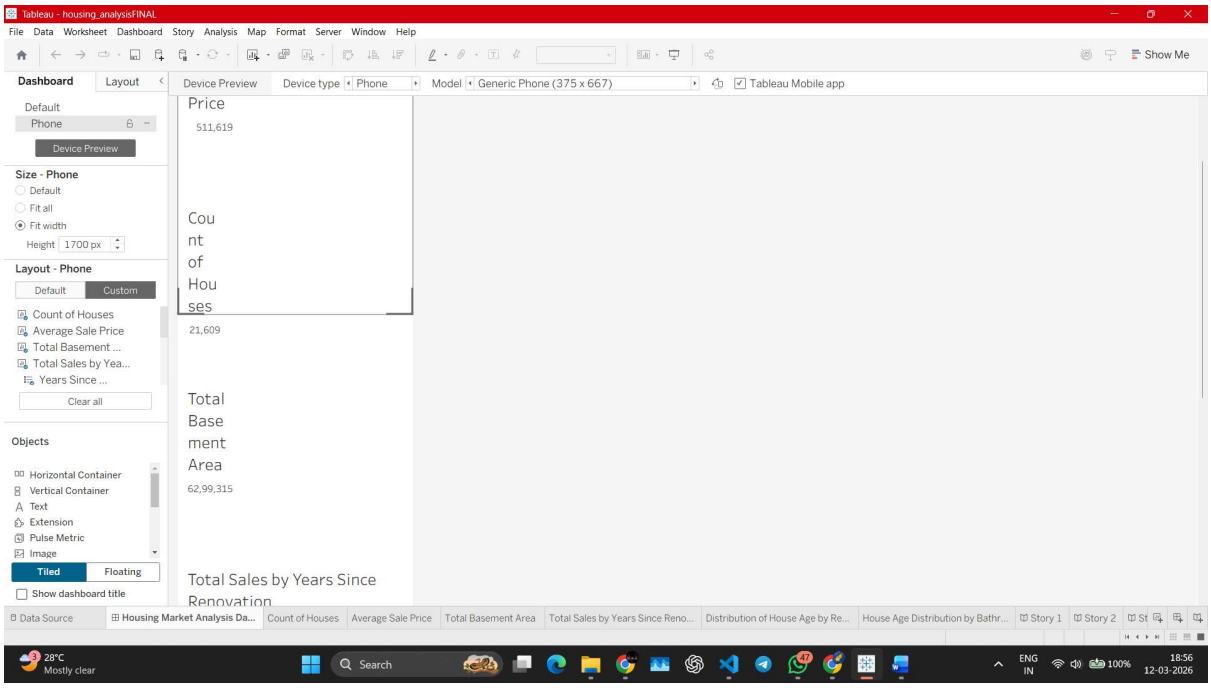
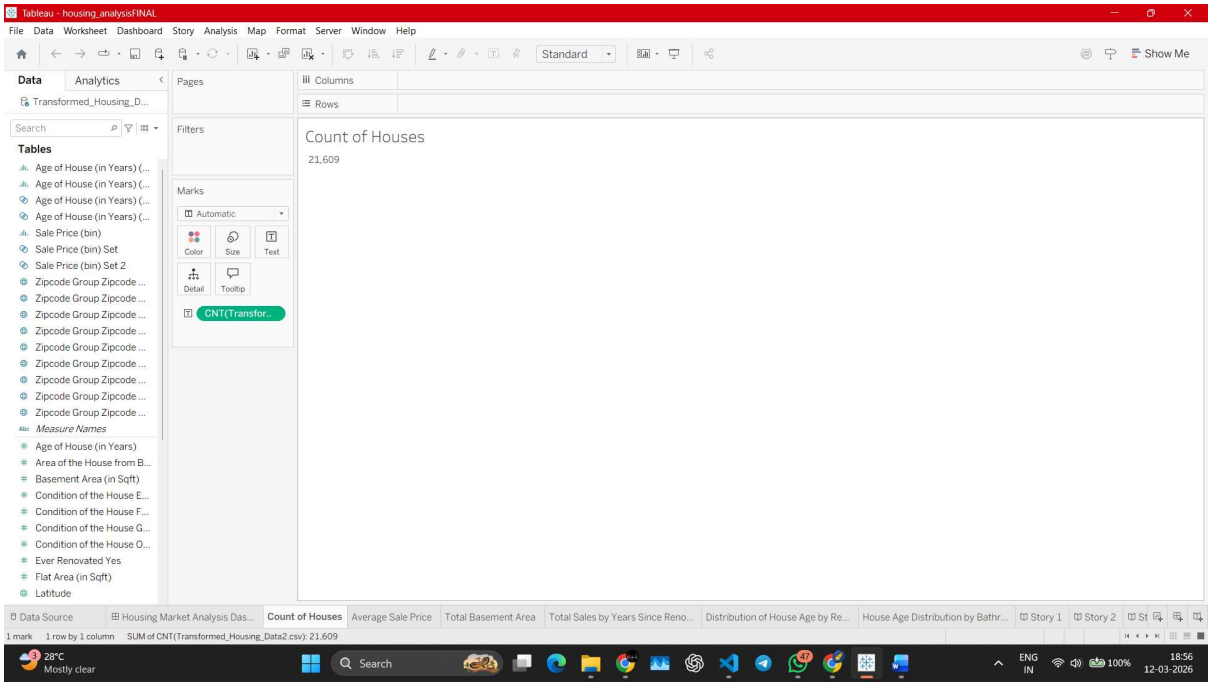
12. Future Improvements

Future improvements may include:

- Adding machine learning models to predict housing prices
- Integrating real-time housing datasets
- Enhancing the web interface with more interactive features
- Deploying the application online

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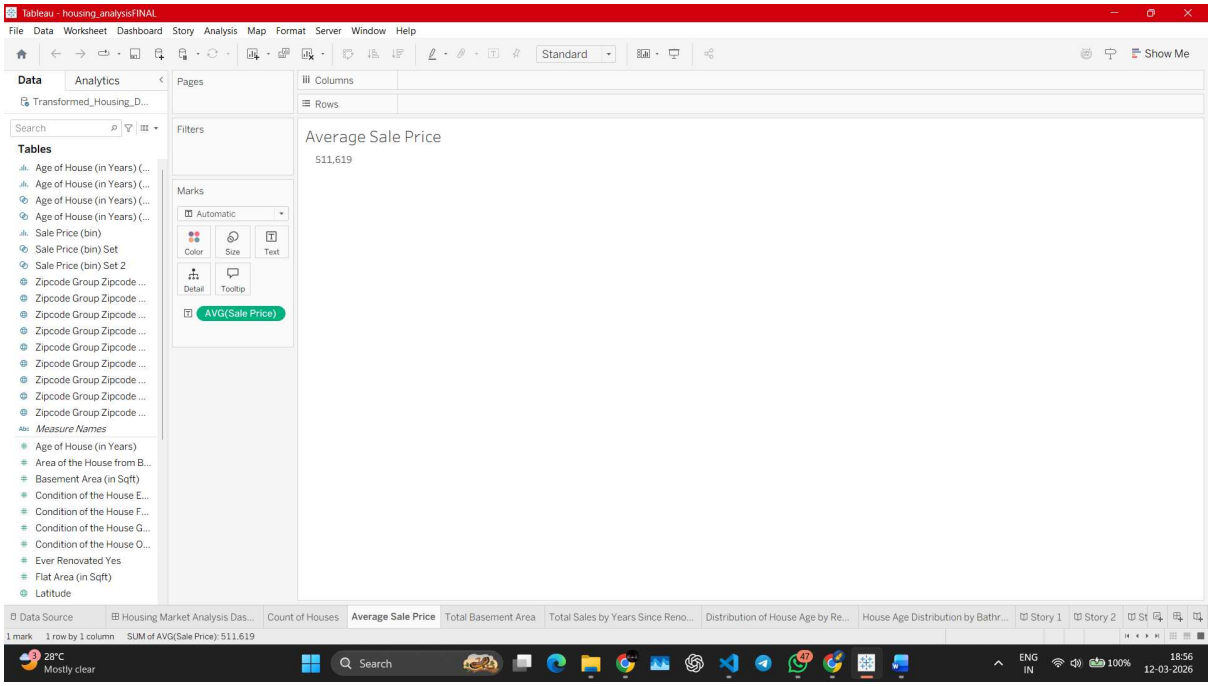
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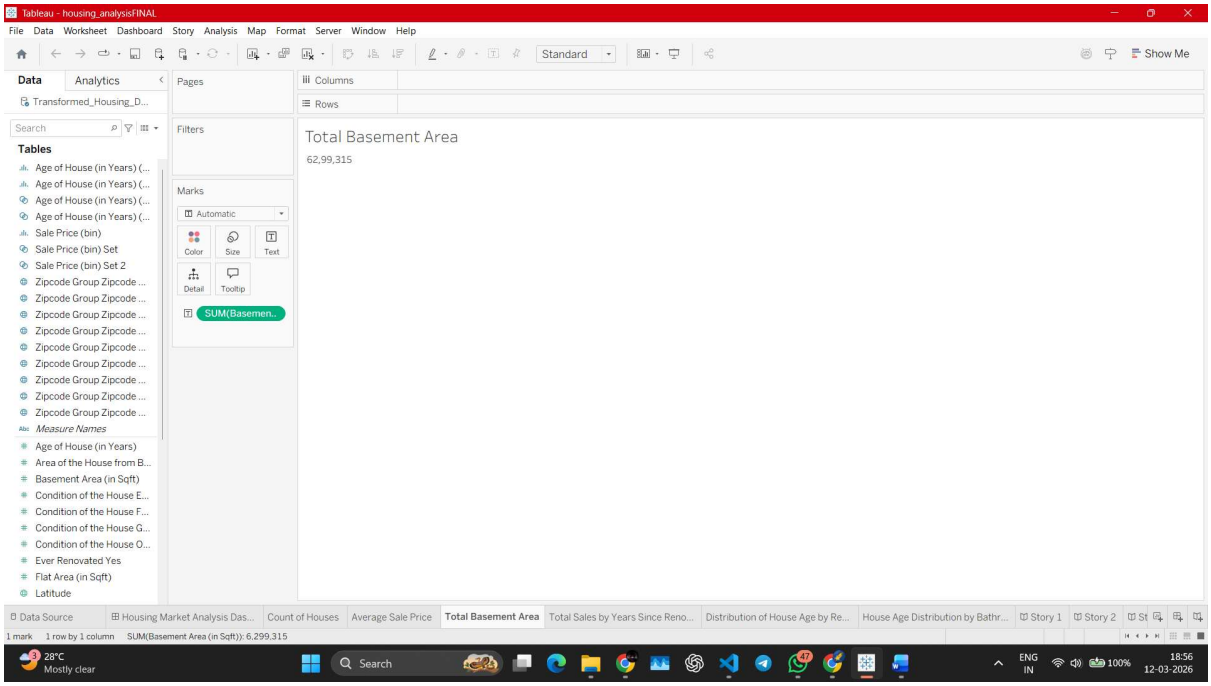
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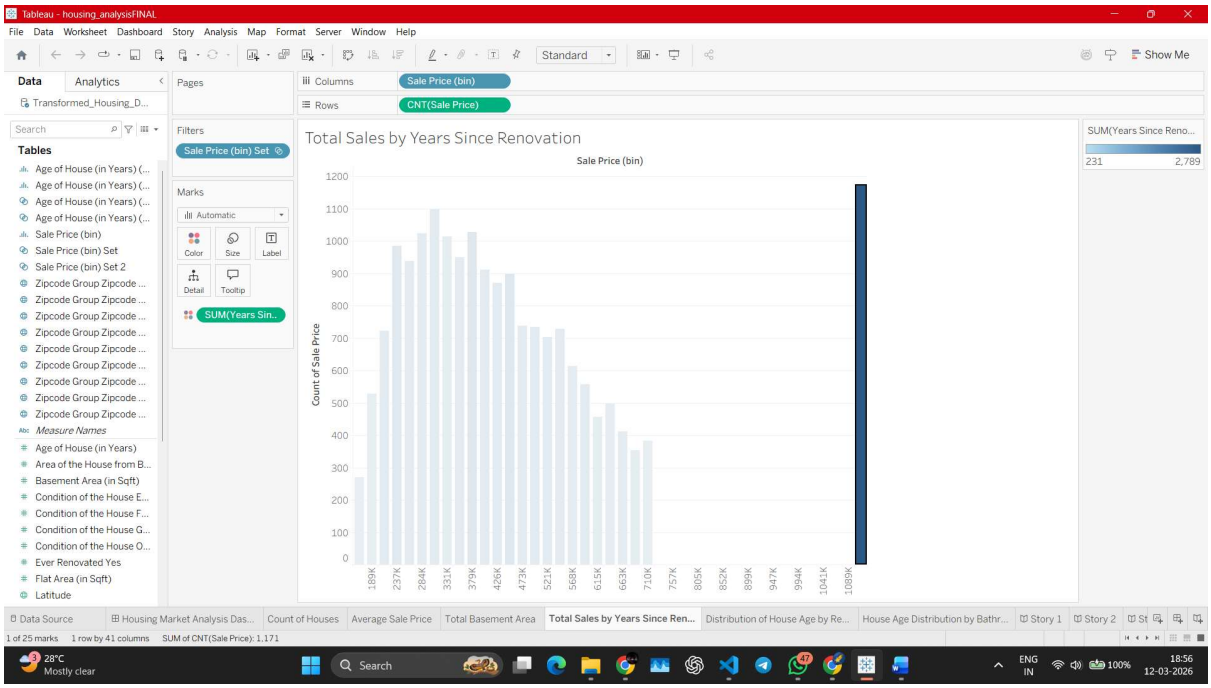
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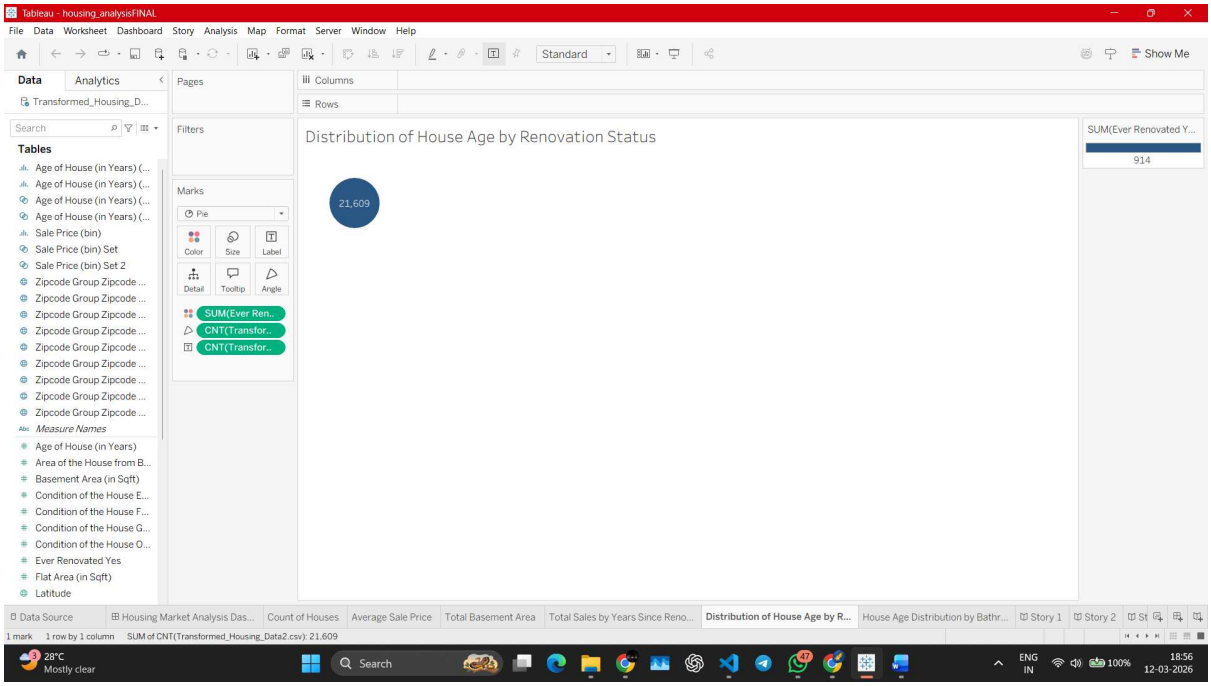
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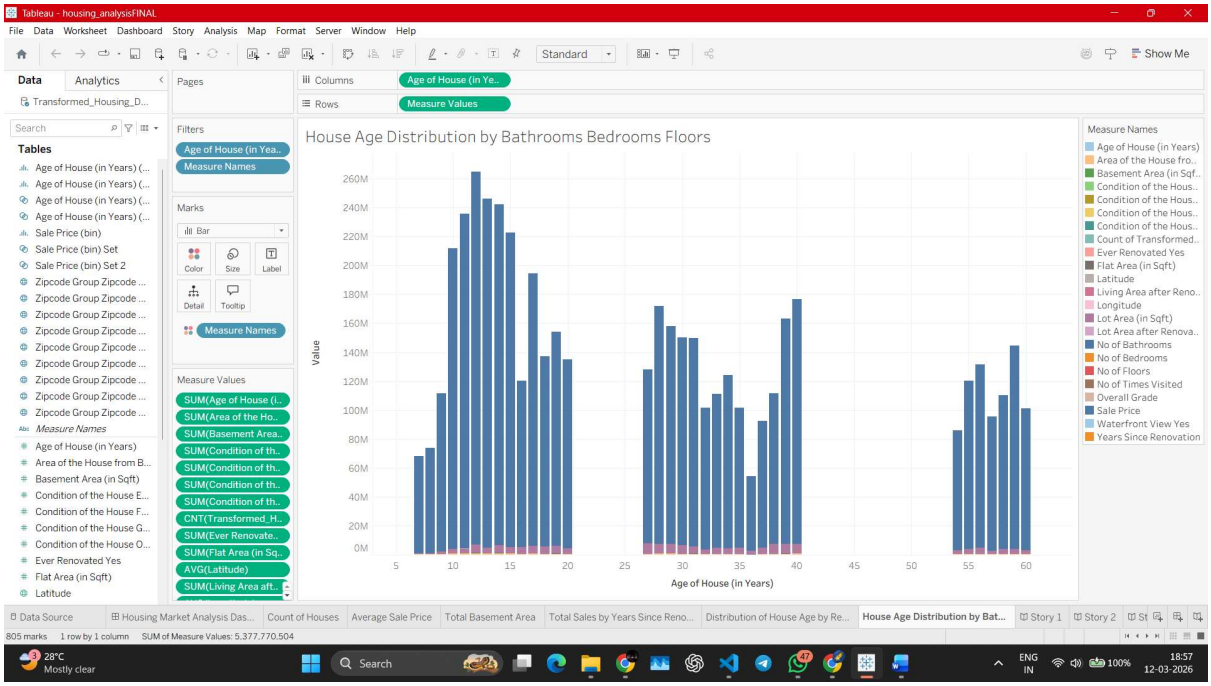
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No of Times Visited
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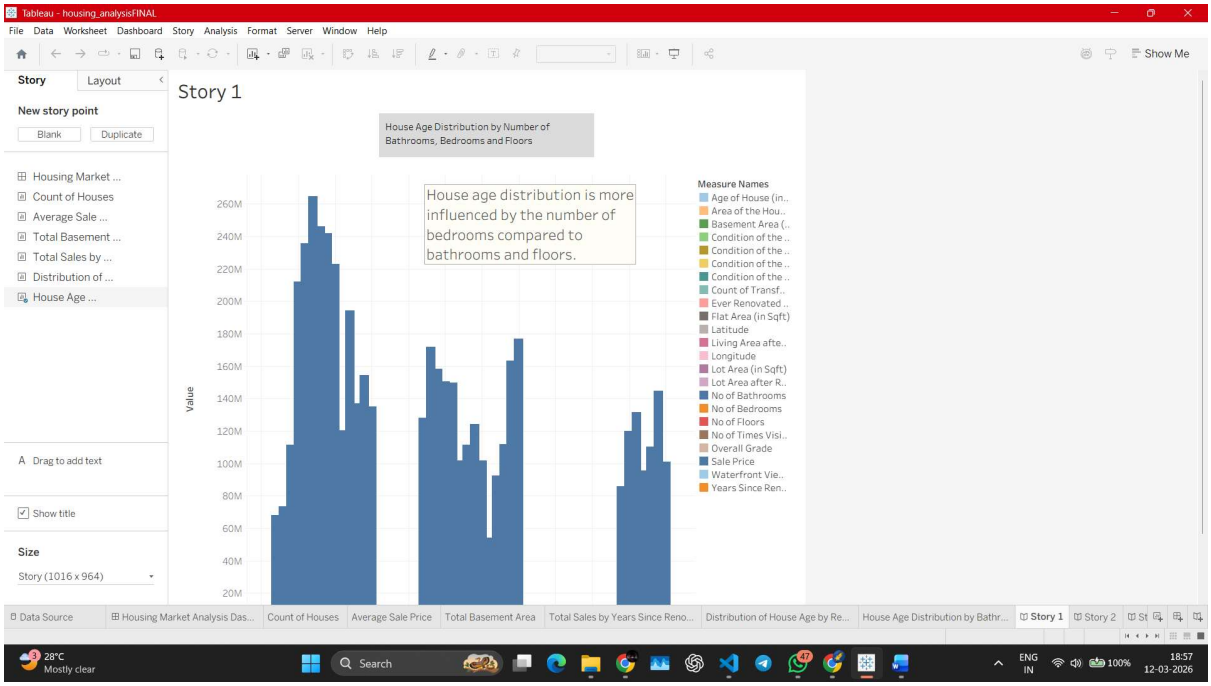
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Total Basement ...
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House Age Distribution by Number of Bathrooms, Bedrooms and Floors
House age distribution is more influenced by the number of bedrooms compared to bathrooms and floors.
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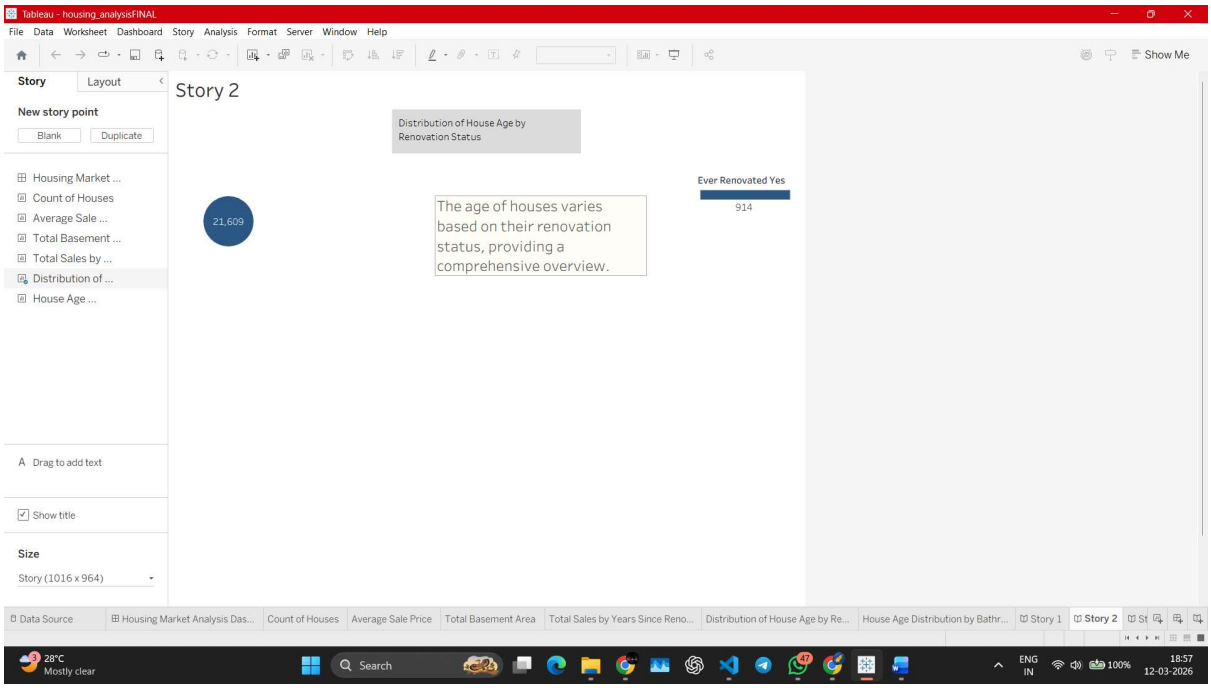
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Story 2
Distribution of House Age by Renovation Status
21,609
The age of houses varies based on their renovation status, providing a comprehensive overview.
Ever Renovated Yes
914
Data Source
Housing Market Analysis Das...
Count of Houses
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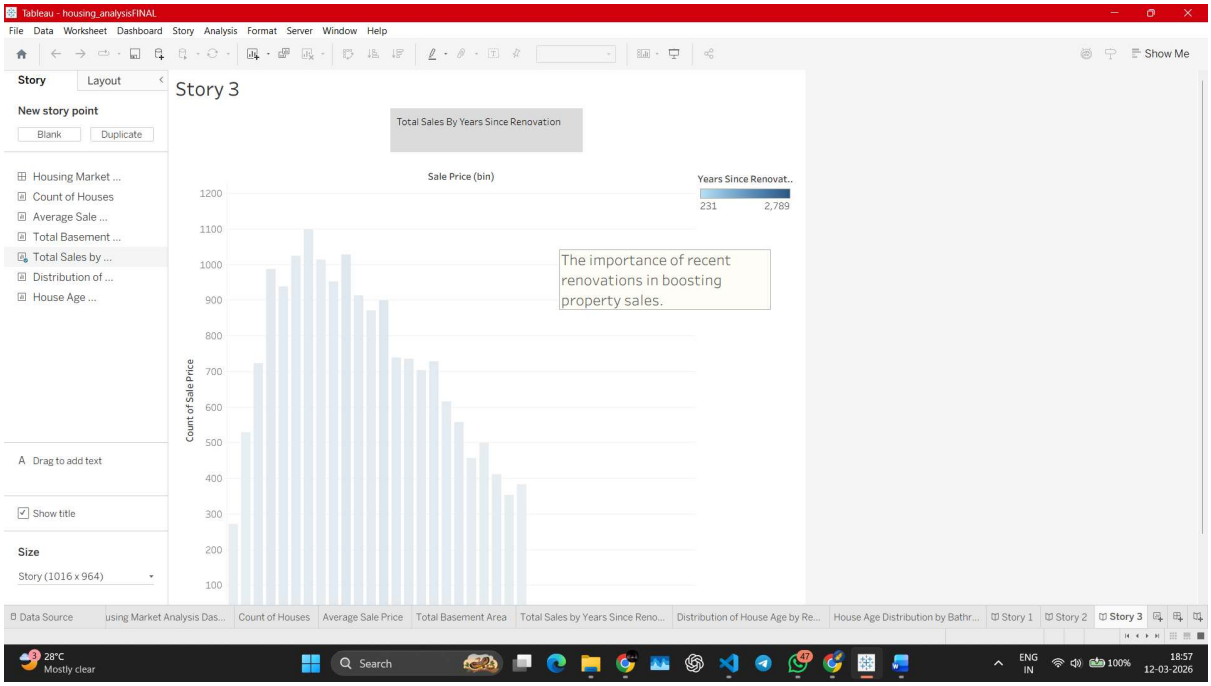
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Story (1016 x 964)
Story 3
Total Sales By Years Since Renovation
Sale Price (bin)
Years Since Renovat..
231
2,789
The importance of recent renovations in boosting property sales.
Count of Sale Price
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